

Notes: Admati and Pfleiderer (RFS, 2000)
Forcing Firms to Talk: Financial Disclosure Regulation and Externalities

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1 Big picture

- **Research question.** Should firms be forced or encouraged to disclose information when disclosure is costly and firms can choose disclosure voluntarily?
- **Core puzzle.** If disclosure improves market liquidity, valuation, and firms' access to transactions, why do firms not voluntarily disclose the socially desirable amount?
- **Main idea.** Disclosure by one firm may help investors value other firms when firm values are correlated. Because the disclosing firm does not internalize this information spillover, voluntary disclosure can be socially inefficient.
- **Setting.** Firms choose the precision of truthful disclosed signals. Precision is costly. Information reduces adverse selection and makes value-enhancing transactions more likely.
- **Single-firm benchmark.** With one isolated firm, the firm chooses the socially optimal disclosure precision because it captures all gains from the transaction.
- **Multiple-firm problem.** With correlated firm values, one firm's disclosure improves information about other firms. Voluntary Nash equilibria may involve too little disclosure, too much disclosure, multiple equilibria, or no pure-strategy equilibrium.
- **Regulation result.** A minimum precision requirement or a subsidy for disclosure can improve welfare, but either can also fail or reduce welfare. Neither regulatory tool dominates.
- **Takeaway.** Disclosure externalities provide a coherent rationale for mandatory disclosure, but the paper warns against the simple view that uniformly "more disclosure" is always socially better.

2 Incremental contribution of the paper

- **From unraveling to costly precision choice.** Earlier unraveling arguments suggest that informed parties voluntarily reveal information because nondisclosure is bad news. This paper emphasizes that disclosure precision is costly and endogenous.
- **Externality-based rationale for regulation.** The paper gives a formal rationale for disclosure regulation based on informational spillovers across firms with correlated values.
- **Precision rather than binary disclosure.** Disclosure is not just yes/no. Firms choose how precise a signal to disclose, and the cost rises with precision.
- **Nonconcavity and discontinuity.** The single-firm disclosure problem has economies of scale in information: low-precision information may be nearly worthless, while sufficiently precise information can be valuable. This generates discontinuous disclosure policies.
- **Game-theoretic disclosure policy.** The paper studies a voluntary disclosure game in which firms choose precision simultaneously and disclosures affect other firms' values through correlated fundamentals.
- **Regulatory ambiguity.** The paper goes beyond "regulation can help" by showing that uniform mandates and subsidies are parameter-sensitive and can be harmful.

- **Targeting insight.** If firms differ, firm-specific or type-specific disclosure standards may be more effective than a uniform standard, but they require information that regulators may not have.

3 Model primitives and economic environment

3.1 One-firm environment

The firm has a retention value $\tilde{v}$, known to current owners but not credibly revealable except through a noisy disclosure technology. A value-increasing transaction is available. If it occurs, firm value rises from $\tilde{v}$ to $\tilde{v} + \delta$, where $\delta > 0$ is the known gain from the transaction.

- Outside investors initially believe $\tilde{v}$ is normally distributed with variance $1/\omega$; ω is prior precision.
- The firm can disclose a signal

$$\tilde{s} = \tilde{v} + \tilde{\eta},$$

where $\tilde{\eta}$ is independent normal noise with variance $1/h$.

- The firm chooses signal precision $h \geq 0$ before knowing the realized value used in the sale decision.
- Disclosure cost is linear in precision: γh , with $\gamma > 0$.

3.2 Adverse selection and transaction pricing

If the firm sells, outside investors infer that current owners are willing to sell. Let p be the transaction price conditional on the sale occurring. Owners sell only if $\tilde{v} \leq p$. The price satisfies

$$p = E(\tilde{v} \mid \tilde{s}, \tilde{v} \leq p) + \delta.$$

- Imperfect disclosure generates adverse selection because high-value firms may withdraw from sale.
- The value of disclosure is that it raises the probability that the value-enhancing transaction occurs.
- With $\delta = 0$, there is a complete market failure under imperfect disclosure because there are no gains from trade to overcome adverse selection.

3.3 Posterior precision and benefit of disclosure

Let $z = \omega + h$ denote total posterior precision after disclosure. The firm chooses $z \geq \omega$ to maximize

$$B(z) - \gamma(z - \omega),$$

where $B(z)$ is the expected transaction benefit from posterior precision z .

The benefit function is increasing in z , but not globally concave. It is convex near zero precision and concave for large precision. This nonconcavity is the source of threshold effects in disclosure.

3.4 Multiple-firm environment

For two firms, retention values $\tilde{v}_1$ and $\tilde{v}_2$ are jointly normal and correlated with coefficient ρ . Firm i can disclose

$$\tilde{s}_i = \tilde{v}_i + \tilde{\eta}_i$$

with precision h_i and cost $\gamma_i h_i$.

- Both signals are publicly observed by investors.
- Each firm chooses precision before observing its realization.
- Each firm's disclosed signal helps value the other firm when $\rho \neq 0$.
- This creates an information externality: the private benefit of disclosure differs from the social benefit.

4 Methodology and theoretical specifications

4.1 Single-firm optimal disclosure

The one-firm problem reduces to a precision-choice problem. The key object is the benefit function $B(z)$. The optimal signal precision is characterized by a tangency between $B(z)$ and the linear cost schedule.

- Lemma 1 shows that higher precision increases the probability that the transaction occurs.
- Proposition 1 shows that the marginal benefit of precision goes to zero as $z \rightarrow 0$; since the benefit function is increasing, it is convex for sufficiently low precision.
- Proposition 2 characterizes the optimal policy: disclose enough to jump to a target precision z^* only for an intermediate range of prior precision; otherwise disclose nothing.

The key intuition is that a small amount of information may not change the adverse-selection problem enough to make high-value firms willing to transact. Once information is sufficiently precise, however, it can sharply increase the probability of trade.

4.2 Reaction functions in the disclosure game

When firm j discloses with precision h_j , this improves investors' information about firm i . The effective prior precision for firm i becomes

$$g_i(h_j) = \frac{\omega_i(\omega_j + h_j)}{\omega_j + h_j(1 - \rho^2)}.$$

Firm i 's posterior precision is $h_i + g_i(h_j)$.

Applying the single-firm solution to this effective prior gives firm i 's reaction function:

$$h_i^r(h_j) = \begin{cases} z_i^* - g_i(h_j), & \text{if } \bar{\omega}_i < g_i(h_j) < z_i^*, \\ 0, & \text{otherwise.} \end{cases}$$

A Nash equilibrium is a pair (h_1^N, h_2^N) such that $h_i^r(h_j^N) = h_i^N$ for both firms.

4.3 Equilibrium possibilities

- With symmetric firms, there is either one or two pure-strategy Nash equilibria.
- If there are two equilibria, one equilibrium involves no disclosure by either firm, while another involves positive disclosure by both firms.
- With asymmetric firms, pure-strategy Nash equilibria need not exist.
- Discontinuous reaction functions are central: small changes in the effective prior precision can cause a jump from no disclosure to positive disclosure.

4.4 Social welfare

The social planner values the total impact of disclosures on all firms, not just the private value to the disclosing firm. When values are correlated, a firm's disclosure helps investors value other firms. Therefore,

private marginal benefit of disclosure $\geq$ social marginal benefit of disclosure.

- If firms' values are uncorrelated, the externality disappears and the private outcome can coincide with the social optimum.
- If firms' values are sufficiently correlated, the Nash equilibrium can underprovide disclosure relative to the social optimum.
- Overinvestment can also occur because a firm may choose precision in response to information produced by others in a way that is not socially aligned.
- The welfare loss generally becomes more important when correlations are stronger or the number of firms is larger.

4.5 Regulation instruments

The paper studies two broad regulatory instruments.

1. **Minimum precision regulation:** firms are required to disclose signals with precision at least $\underline{h}$.
2. **Disclosure subsidies:** firms face a lower perceived disclosure cost, for example $\gamma - y$ instead of γ .

Both instruments can improve welfare by inducing more disclosure or selecting a better equilibrium. But both can also fail because the optimal intervention depends on which firm should disclose and how much. A uniform rule can force the wrong firms to disclose or can increase disclosure when the problem is not underdisclosure.

5 Interpretation and important assumptions

5.1 What the paper is and is not about

The paper is about the amount and precision of truthful disclosure. It is not primarily about antifraud regulation or penalties for false statements. Truthful disclosure is assumed.

5.2 Why disclosure can matter

Disclosure matters because asymmetric information blocks efficient transactions. If outside investors do not know firm value, high-value current owners may refuse to sell at a pooled price. More precise signals reduce this adverse selection and increase the chance that socially valuable transactions occur.

5.3 Why single-firm disclosure need not justify regulation

In the single-firm benchmark, the firm internalizes the full value of disclosure because it captures all transaction surplus. Regulation becomes interesting only with multiple correlated firms or if firms capture only part of the surplus.

5.4 Why correlated values matter

Correlation turns disclosure into a public input. A signal about firm j also informs investors about firm i . Private firms do not fully count these cross-firm benefits when choosing precision.

5.5 Why uniform regulation is hard

The regulator may not know which firms generate the largest spillovers or which firms have the lowest disclosure costs. A uniform rule may be too crude when firms differ by industry, size, correlation, cost, or transaction gain.

6 Figures

6.1 Figure 1: Single-firm benefit function and threshold disclosure

Read:

- Figure 1 shows the general shape of the disclosure benefit function $B(z)$ and the linear disclosure cost schedule.
- The benefit function is convex at low precision and concave at higher precision.
- The optimal policy has three regions: no disclosure at low prior precision, a jump to z^* at intermediate prior precision, and no extra disclosure when prior precision is already high.

Economic message: Disclosure can have scale economies. Low precision may be nearly useless, while sufficiently precise disclosure can be valuable enough to justify a discrete jump in information production.

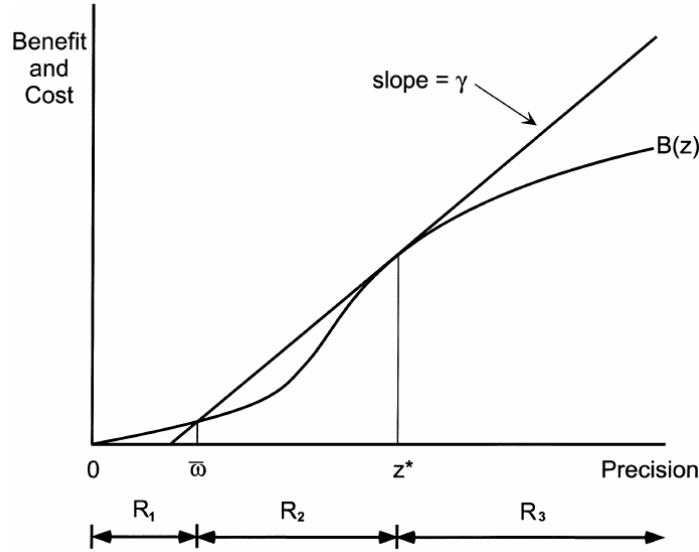


Figure 1: Benefit and cost of disclosure as a function of total precision.

6.2 Figures 2A and 2B: Discontinuous optimal disclosure

Read Figure 2A:

- The optimal disclosure precision is plotted against prior precision ω .
- For very low prior precision, the firm discloses nothing.
- At a threshold, the firm jumps to a high disclosure precision.
- As prior precision rises further, the additional signal precision falls and eventually reaches zero.

Read Figure 2B:

- The optimal disclosure precision is plotted against the transaction gain δ .
- Disclosure also jumps discontinuously as the transaction gain crosses a threshold.
- Larger gains from trade make disclosure more valuable, but the relation is not smooth.

Economic message: The model does not predict gradual disclosure adjustment. Optimal disclosure can jump sharply when fundamentals make information just valuable enough.

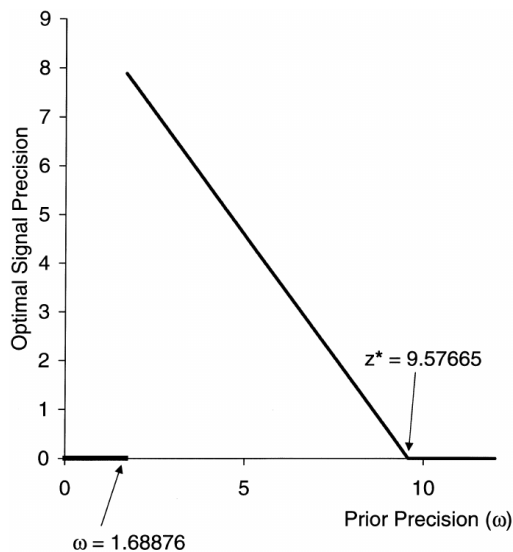


Figure 2A: Optimal precision as a function of prior precision.

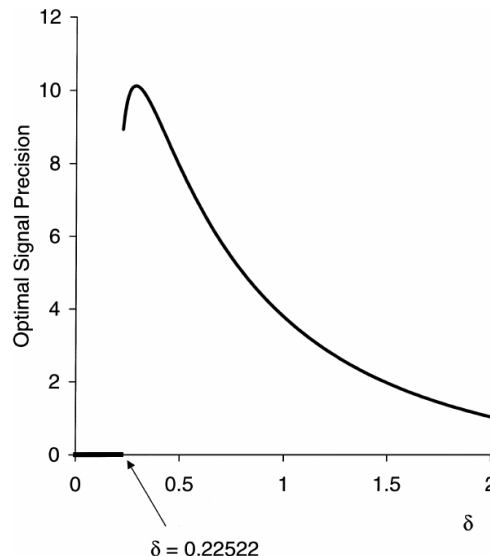


Figure 2B: Optimal precision as a function of transaction gains.

6.3 Figures 3 and 4: Reaction functions and equilibrium existence

Read Figure 3:

- Figure 3 plots reaction functions for two identical firms.
- The game has two Nash equilibria: no disclosure by both firms, and positive disclosure by both firms.
- Discontinuous reaction functions generate equilibrium multiplicity.

Read Figure 4:

- Figure 4 gives an asymmetric-firm example with no pure-strategy Nash equilibrium.
- One reaction function is continuous while the other is discontinuous.
- The reaction functions do not intersect.

Economic message: Voluntary disclosure games can be unstable. Correlated values and discontinuous private incentives can create multiple equilibria or eliminate pure equilibria.

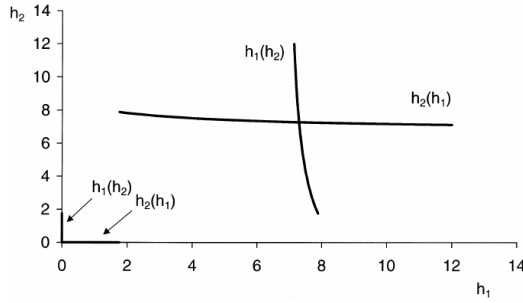


Figure 3: Reaction functions with two identical firms.

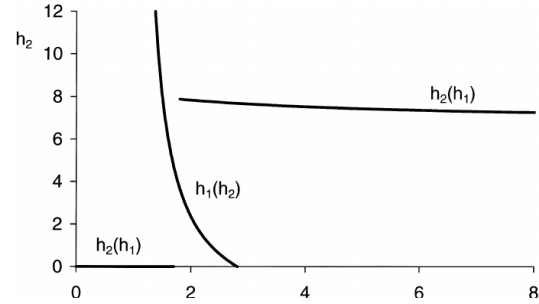


Figure 4: Example with no pure-strategy Nash equilibrium.

6.4 Figures 5A and 5B: Social optimum versus Nash equilibrium

Read Figure 5A:

- The socially optimal precision and Nash precision are plotted against the correlation ρ between firm values.
- For low correlation, the no-disclosure Nash equilibrium can coincide with the social optimum.
- For higher correlation, the social optimum jumps to positive disclosure and lies above the positive-disclosure Nash equilibrium.

Read Figure 5B:

- Attained social value is plotted for the same example.
- The social optimum dominates Nash outcomes when correlation is high.
- The no-disclosure Nash equilibrium becomes especially inefficient at high correlation.

Economic message: The disclosure externality becomes important when firms' values are highly correlated. The more a firm's signal helps value another firm, the more voluntary disclosure can fall short of the social optimum.

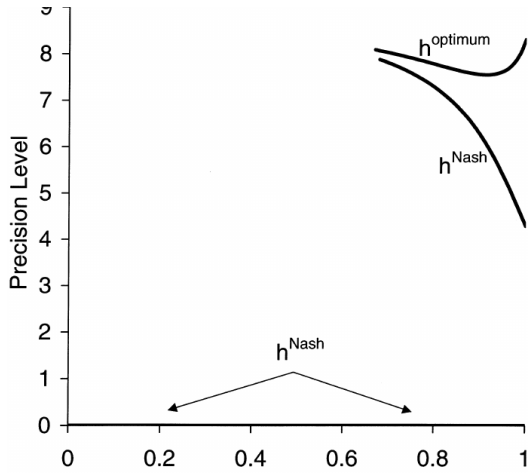


Figure 5A: Precision levels as correlation changes.

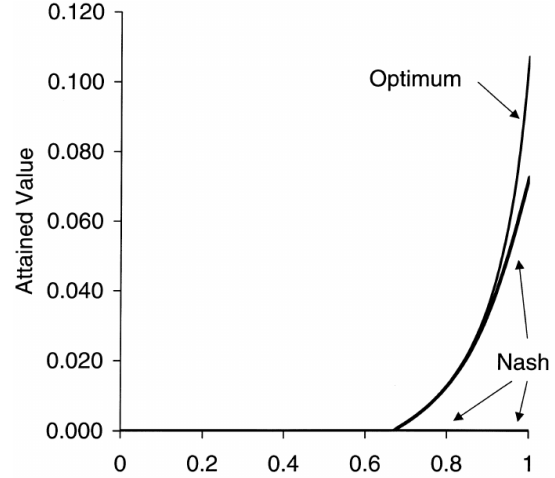


Figure 5B: Attained value as correlation changes.

6.5 Figures 6A and 6B: Many firms and free riding

Read Figure 6A:

- The socially optimal precision is plotted as the number of firms rises.
- The Nash equilibrium has zero disclosure for all N in the example.
- The social optimum becomes positive once the number of firms is sufficiently large.

Read Figure 6B:

- Social welfare in the optimum rises relative to the Nash outcome as the number of firms grows.
- Each disclosure becomes valuable to many other firms, but the disclosing firm does not internalize these spillovers.

Economic message: With many correlated firms, the public-good aspect of disclosure becomes stronger. Free riding can make the voluntary outcome increasingly inefficient.

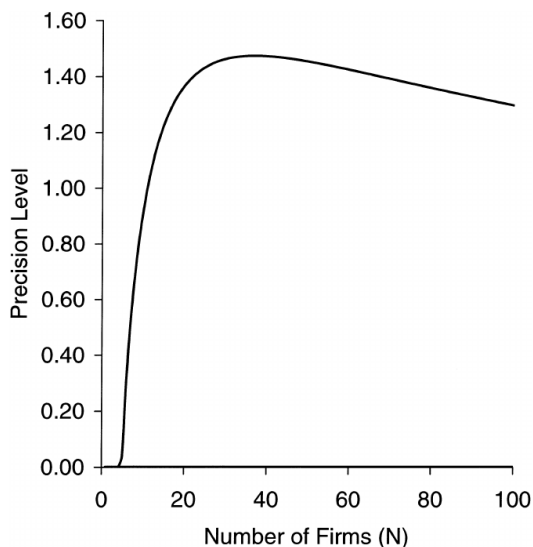


Figure 6A: Social precision as the number of firms grows.

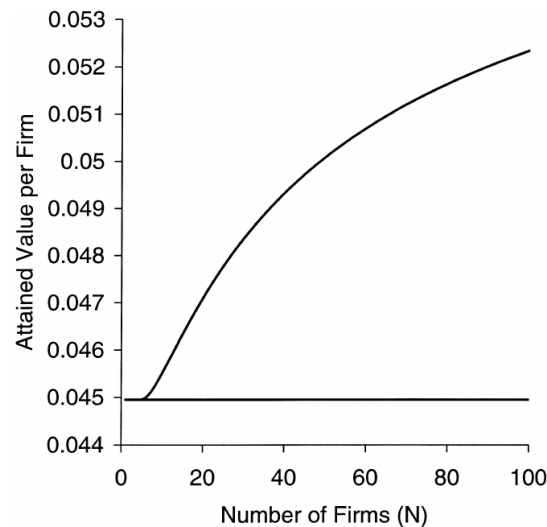


Figure 6B

Figure 6B: Welfare loss from free riding as N grows.

6.6 Figures 7A and 7B: Asymmetric firms

Read Figure 7A:

- The example has two asymmetric firms with different gains and disclosure costs.
- The Nash equilibrium is constant across correlation levels in the example.
- The social optimum changes with correlation and can call for very different precision choices across firms.

Read Figure 7B:

- Social welfare under the optimum rises above the Nash outcome as correlation increases.
- The inefficiency becomes larger at high correlation.

Economic message: When firms differ, the regulator's problem is not just how much disclosure to require, but also which firms should produce the information.

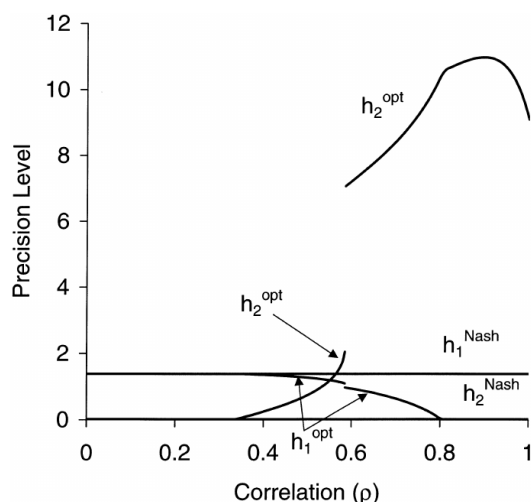


Figure 7A: Nash and social precision with asymmetric firms.

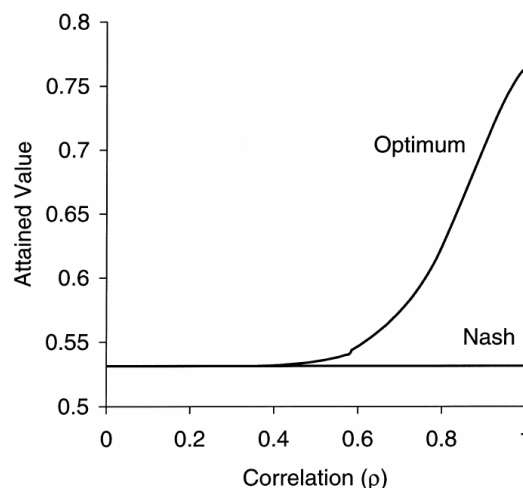


Figure 7B: Welfare with asymmetric firms.

6.7 Figures 8A and 8B: Minimum precision can improve welfare

Read Figure 8A:

- A uniform minimum precision requirement changes the Nash disclosure choices.
- At low mandated precision, the constraint binds on one firm but not the other.
- As the requirement rises, both firms may be forced to disclose at the minimum level.

Read Figure 8B:

- In this example, some minimum precision requirements raise social welfare above the unregulated Nash equilibrium.
- The welfare-improving range is limited; the optimum is not achieved by simply making the requirement very high.

Economic message: Mandatory disclosure can help when the unregulated equilibrium involves the wrong level or source of information, but the mandate has to be calibrated.

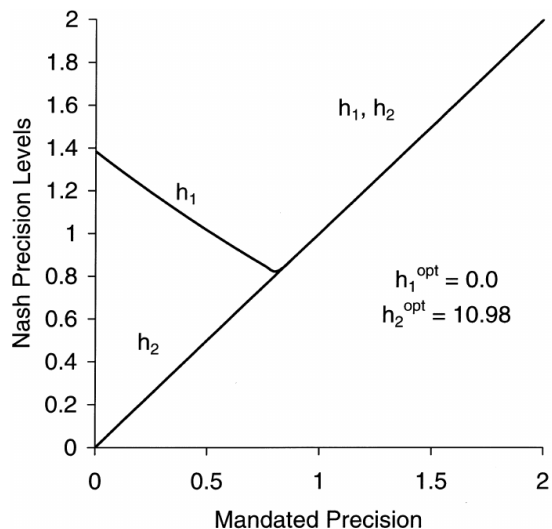


Figure 8A: Nash precision under minimum precision regulation.

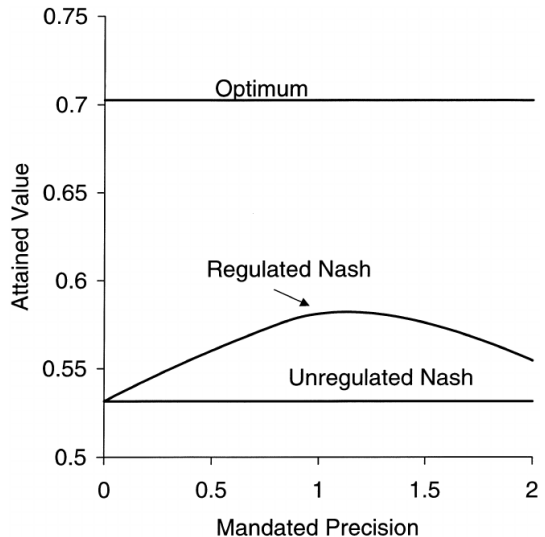


Figure 8B: Welfare under minimum precision regulation.

6.8 Figures 9A and 9B: Minimum precision can reduce welfare

Read Figure 9A:

- The social optimum has one firm disclosing a high-precision signal while the other firm discloses nothing.
- A uniform minimum precision requirement forces both firms to disclose as the mandate rises.
- This can push disclosure onto a high-cost or socially inappropriate firm.

Read Figure 9B:

- In this example, any positive minimum precision lowers social welfare relative to no regulation.
- The mandate cannot fix the underlying inefficiency because it is not targeted by firm type.

Economic message: Uniform mandates can be harmful. The fact that the unregulated equilibrium is inefficient does not guarantee that a uniform minimum disclosure rule improves welfare.

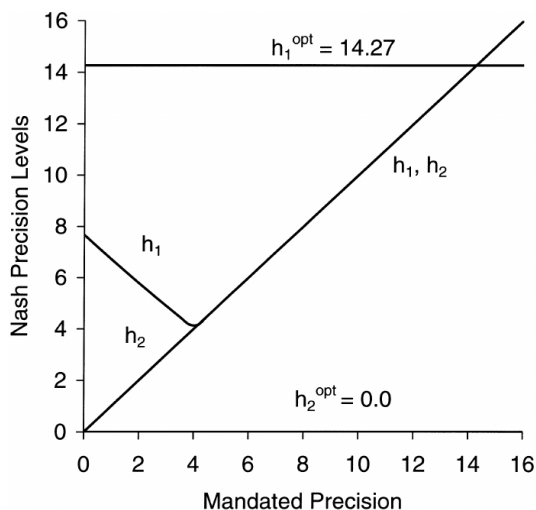


Figure 9A

Figure 9A: A case where minimum precision forces inefficient disclosure.

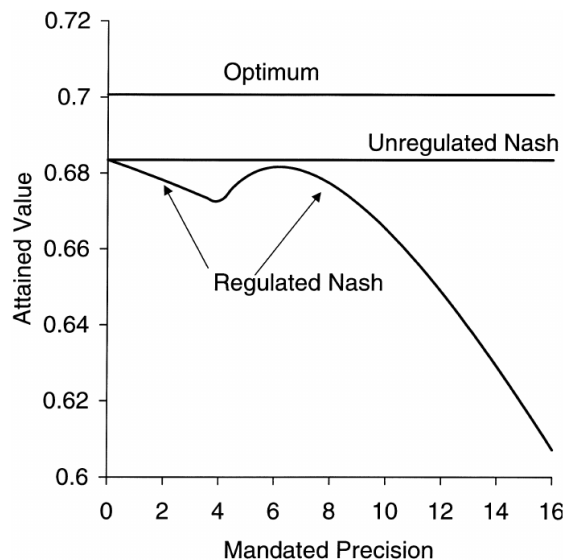


Figure 9B: Welfare can fall under minimum precision regulation.

6.9 Figures 10A and 10B: Subsidies for disclosure

Read Figure 10A:

- A uniform subsidy reduces the perceived cost of disclosure for both firms.
- Subsidies change equilibrium precision choices, but not necessarily toward the social optimum.
- The response can be nonlinear and discontinuous.

Read Figure 10B:

- A small subsidy can improve welfare relative to the unregulated Nash outcome in this example.
- Larger or intermediate subsidies may reduce welfare, depending on how they distort firm choices.
- Subsidies do not dominate minimum precision regulation, and minimum precision regulation does not dominate subsidies.

Economic message: Subsidies are another way to encourage disclosure, but they are also blunt. Like mandates, they require knowledge of the environment to be reliably welfare-improving.

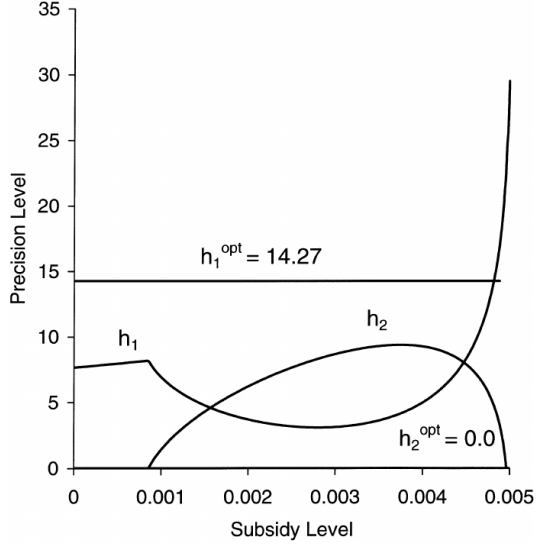


Figure 10A: Equilibrium precision under disclosure subsidies.

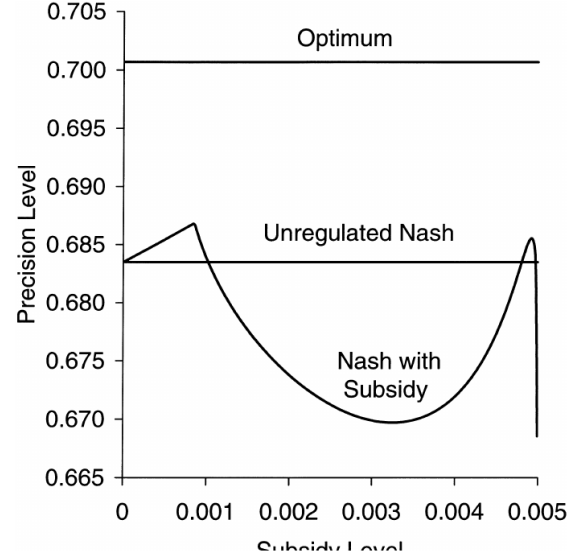


Figure 10B: Welfare under disclosure subsidies.

7 Short summary

- The paper builds a model of costly voluntary disclosure with endogenous signal precision.
- Disclosure has private value because it reduces adverse selection and makes value-enhancing transactions more likely.
- In the one-firm case, the firm chooses socially optimal disclosure precision under the maintained surplus-capture assumption.
- The benefit of disclosure is nonconcave: low precision can be nearly useless, while enough precision can be valuable.
- This nonconcavity generates discontinuities in optimal disclosure policies.
- With multiple firms, correlated values create disclosure externalities because one firm's signal helps value other firms.
- Voluntary disclosure Nash equilibria can be inefficient, and the inefficiency is more pronounced when correlations are high or many firms benefit from a disclosure.
- Minimum disclosure requirements can improve welfare in some cases, especially by eliminating a bad no-disclosure equilibrium.
- Minimum requirements can also reduce welfare if they force the wrong firms to disclose or require too much disclosure.
- Disclosure subsidies can also improve or reduce welfare; neither subsidies nor mandates dominate in all cases.
- The main policy lesson is that disclosure regulation may be justified by externalities, but good regulation must be carefully targeted.

8 Conclusion

8.1 Core conclusion

Admati and Pfleiderer show that disclosure regulation can be justified even when firms voluntarily choose disclosure and disclosed information is truthful. The key reason is that disclosure has social value beyond its private value when firm values are correlated. A firm's signal helps investors value other firms, but the disclosing firm does not fully internalize that benefit.

8.2 Economic intuition

Information is valuable because it mitigates adverse selection. Yet information production can have economies of scale: a little information may not change behavior, while a sufficiently precise signal can unlock a valuable transaction. In multi-firm settings, the same information can be useful across many correlated firms, turning disclosure into a public-good-like input.

8.3 Incremental contribution revisited

The paper moves the disclosure-regulation debate beyond unraveling and beyond single-firm models. It shows that mandatory disclosure can be grounded in cross-firm information transfers, but also that the design of regulation is delicate.

8.4 Implications

Disclosure mandates are most compelling when firm values are highly correlated, when many firms benefit from each disclosure, or when the voluntary equilibrium involves no disclosure. However, uniform regulation may fail when firms differ substantially in disclosure costs, transaction gains, or informational spillovers.

8.5 Limitations

The analysis is theoretical and depends on assumptions such as truthful disclosure, exogenous precision costs, simultaneous precision choices, and risk-neutral pricing. The paper does not estimate optimal disclosure rules empirically, and the regulator's information problem is central.

8.6 Takeaway

The enduring takeaway is that financial disclosure can create external benefits. Regulation can force firms to talk, but forcing all firms to talk in the same way is not automatically efficient.